

ber 2021, November 2021, December 2021, and finally January 2022.

If we look at Fig. 5(a) and Fig. 5(b), the daily variations and the monthly average plots for October and November just form a single band. However, for the months of December 2021 and January 2022, as shown in Fig. 5(c) and Fig. 5(d), it can be easily noticed that the average line lies in the middle, while some days form a band higher in magnitude than the average and some days are in another band that is below the monthly average line.

This clearly indicates that for some days, the signal variations were higher than average while for some days these were way below the average. Therefore, simple statistical parameter estimation for each of these days for all four months was carried out. These are minima, maxima, average, standard deviation, and the term delta, which is the difference between maxima and minima.

The following figures illustrate respectively daily averages for the four months. The average values or the standard deviation for December 2021 shows a significant difference and has revealed a shocking fact that the geophysical process associated with this volcano eruption started many weeks before. The process had begun almost a month in advance on 17th December 2021 and was terminated on 14th January 2022. Following this termination, on 15th January 2022, the volcanic eruption physically appeared on the sea-surface.

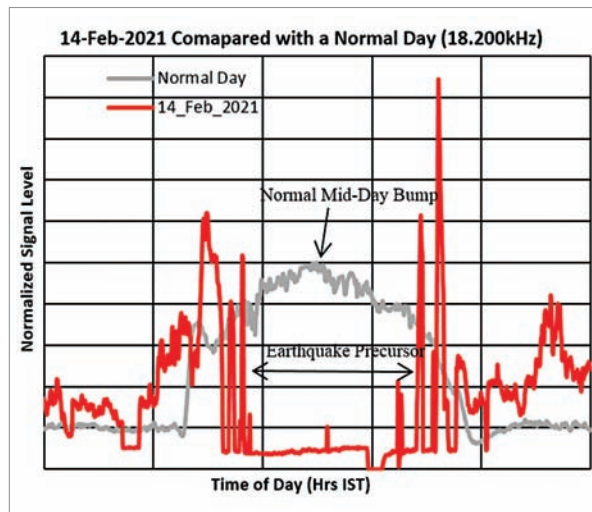


Fig. 7(a): The earthquake precursor obtained in the same 18.200kHz VLF channel on 14th February 2021

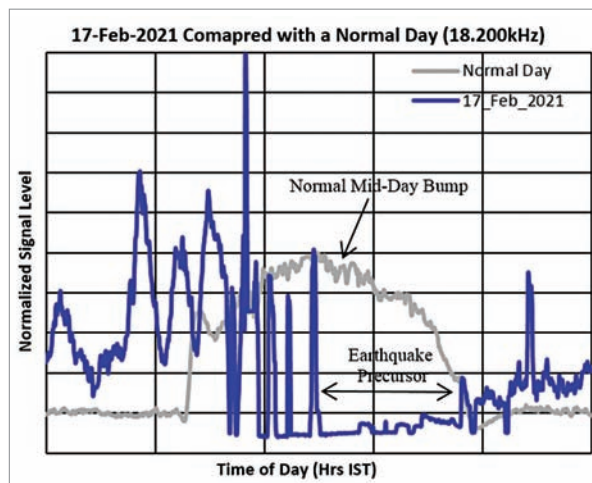


Fig. 7(b): The earthquake precursor obtained in the same 18.200kHz VLF channel on 17th February 2021

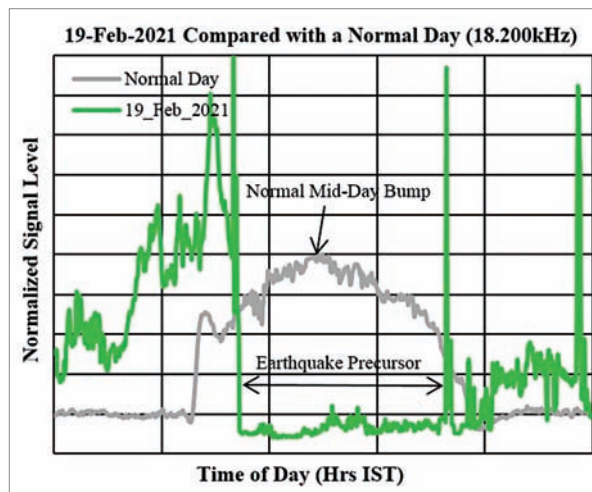


Fig. 7(c): The earthquake precursor obtained in the same 18.200kHz VLF channel on 19th February 2021

Comparison with earthquake precursors

By looking at Fig. 6(c) and Fig. 6(d), it was immediately clear that prior indications, that is, the precursors associated with major earthquakes, have identical shape of the signal strength variations and the same precursors are then used to predict earthquakes.

The following Fig. 7(a), 7(b), and 7(c) illustrate the precursors obtained on 14th, 17th, and 19th February 2021 that were later found to be associated with three major earthquakes of M7.3, M7.4, and M8.1 that took place on 4th March 2021 in the Kermadec Islands and New Zealand in the South Pacific region. Using the first detected precursor on 14th February 2021, the author had predicted the possibility of a M7+ earthquake and had issued an earthquake alert more than 14 days in advance, indicating the possibility of M7+ earthquake events in the South Pacific region, including New Zealand. See Fig. 8.

Notice the clear-cut similarities in the shape and pattern in signal strength changes, both during a volcano eruption as shown in Fig. 6(c) and 6(d) and a major earthquake as shown in Fig. 7(a), 7(b), and 7(c). There is always a drop in the VLF signal level as an indicator of these pre-volcanic or pre-seismic geophysical processes.

Also, notice that in comparison with Fig. 4, which shows a 'normal' curve, the mid-day peak is totally absent from these precursors.